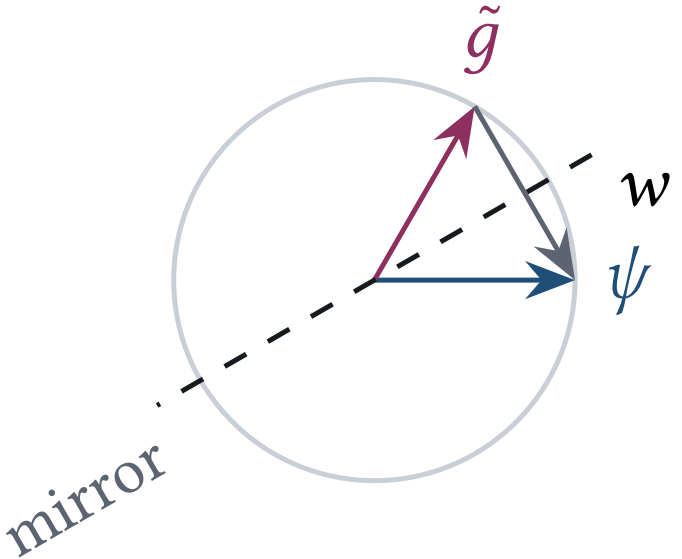


$$R^{\psi} = I - 2\frac{ww^{\dagger}}{\|w\|^2}$$

$R^{\psi} = I$ if $w = 0$.